Wenyin (Knut) Gu

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EDUCATION

- **University of California, Irvine** | Irvine, CA | 2025 - Present

  B.S. in Electrical Engineering (Specialization: Digital Signal Processing)

  GPA: 3.69/4.0 | Dean's List (All Quarters)

- **Cañada College** | Redwood City, CA | 2022 - 2025

  A.S. in Engineering

  GPA: 4.0/4.0 | Valedictorian, *Summa Cum Laude*, Dean's List (All Semesters)

EXPERIENCE

- **Tutor of Mathematics**, *Cañada College* | 2022 - 2024

  Provided one-on-one, drop-in tutoring at the Learning Centre, guiding students through mathematical problems and supporting their development of analytical skills.

- **Fundamentals of Data Science Workshop**, *Stanford University* | Summer 2023

  Completed a 4-week intensive program encompassing Applications of Data Science, Python, Statistics, Mathematical Optimization, Linear Algebra, and Responsible AI.

- **Outstanding Service Contributor (Water Supply Team Lead)**, *Shanghai World Expo* | 2010

  Led a team to conduct comprehensive safety inspections of water supply networks and enforce administrative penalties for water-related violations, successfully safeguarding the drinking water security for over 73 million visitors throughout the 6-month event.

- **Volunteer**, *Special Olympics World Summer Games* | 2007

  Provided dedicated care and round-the-clock support for a football team of 10+ athletes with intellectual disabilities, ensuring their safety and well-being during daily routines and training.

PROJECTS

- **Spatial vs. Temporal Edge Detection Simulation** | *Independent Research Project*

  Developed Python simulations (NumPy/SciPy) comparing 2D spatial convolution (Sobel operators) with temporal high-pass filtering (difference equations) for edge detection.

  Demonstrated event-based temporal filtering achieves superior signal sparsity and efficiency in motion tracking by mathematically suppressing static backgrounds to zero.

- **Music Reactive LED System** | *Electrical Engineering Project*

  Designed a custom microphone preamplifier and 3-band active analog crossover (LM358 Op-Amps) to hardware-separate audio frequencies (Bass, Midrange, Treble).

  Programmed an Arduino non-blocking state machine and an interactive OLED GUI to process multi-channel ADC signals into three real-time LED visualization modes.

SKILLS

- **Software & Programming**: Python, MATLAB, SolidWorks, Onshape, C/C++ (Arduino).
- **Hardware & Electronics**: Analog Circuit Design (Op-Amps, Active Filters), Microcontroller Integration (ADC, I2C), UI System Implementation, Soldering/Perfboard Fabrication.

AWARDS and HONOURS

- IEEE-Eta Kappa Nu (HKN) | 2026
- Phi Theta Kappa | 2023